

EBAU - 2021 EKAINA - [BU]

$$f(x) = e^x$$

$$g(x) = e^{-x}$$

$$h(x) = y = e$$

Bi funtzioen ebaketa puntua.

$$e^x = e^{-x}$$

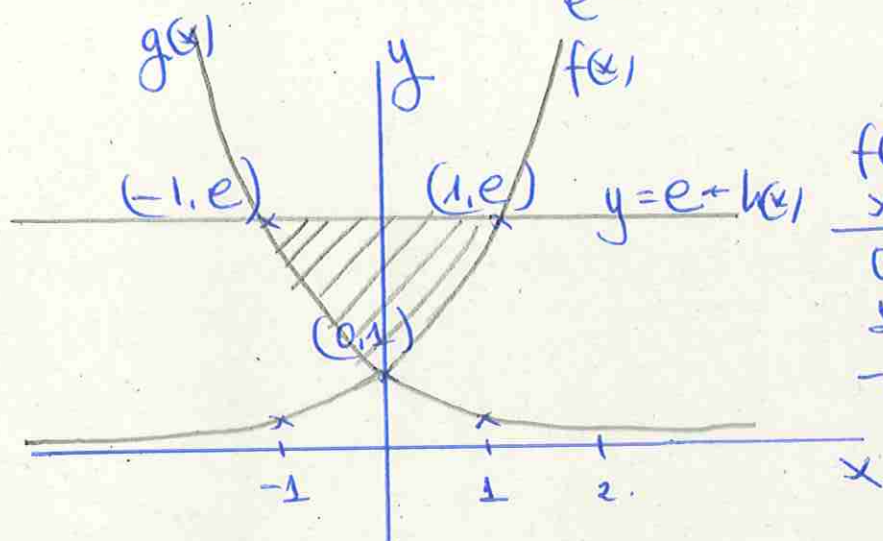
$$\frac{e^x}{e^{-x}} = 1$$

$$e^{x+x} = 1$$

$$2x = 0$$

$$x = 0, y = 1$$

$$P(0, 1)$$



$f(x)$	x	y
	0	1
	1	$e = 2,7$
	-1	$e^{-1} = 0,37$

$g(x)$	x	y
	0	1
	1	$e^{-1} = 0,37$
	-1	$e = 2,7$

$$A = \int_{-1}^0 (h(x) - g(x)) dx + \int_0^1 (h(x) - f(x)) dx$$

$$= 2 \int_{-1}^0 (h(x) - g(x)) dx = \boxed{2 \int_0^1 (h(x) - f(x)) dx}$$

$$A = 2 \int_0^1 (h(x) - f(x)) dx = 2 \int_0^1 (e - e^x) dx =$$

$$= 2 \left[ex - e^x \right]_0^1 = 2 \left[(e - e) - (0 - e^0) \right] =$$

$$= 2 \cdot 1 = \boxed{2}$$

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$$f(x) = x$$

$$g(x) = x/8$$

$$h(x) = 1/x^2$$

Lehen kuadrantea.

Hiru funtzioen adierazpen grafikoa

→ $h(x) = 1/x^2$ Dom = $\mathbb{R} - \{0\}$.
Simetrikoa da.

$$h(x) = \frac{1}{x^2}$$

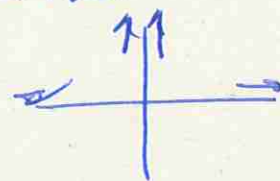
Asintotak - Bertikala $x=0$. $\lim_{x \rightarrow 0^+} \frac{1}{x^2} = +\infty$

Horizontala

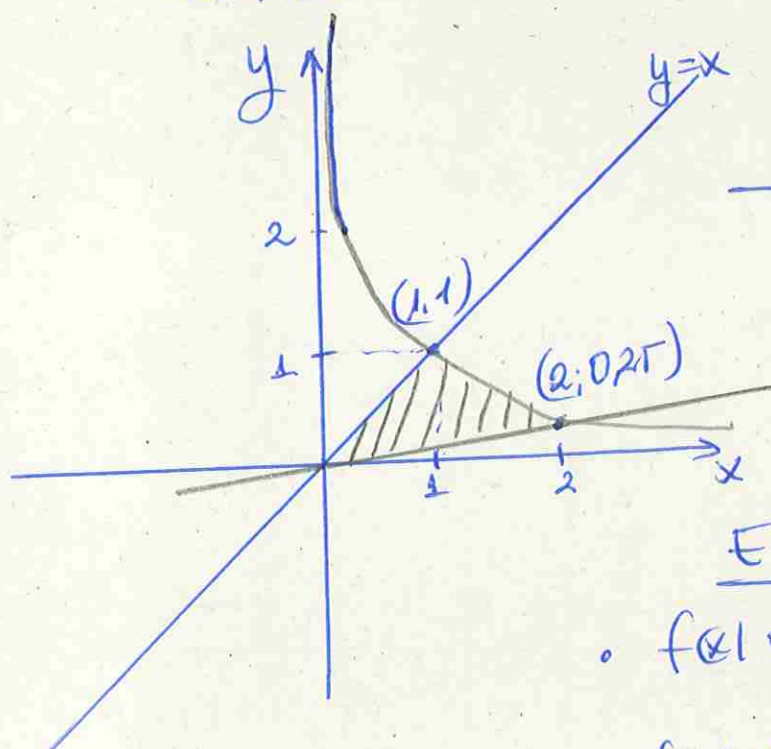
$$\lim_{x \rightarrow \infty} \frac{1}{x^2} = 0 \rightarrow y=0$$

$$x \rightarrow +\infty \quad \frac{1}{+} = 0^+ > 0 \text{ positibua}$$

$$x \rightarrow -\infty \quad \frac{1}{+} = 0^+ > 0 \text{ positibua}$$



x	y
1	1
2	0,25
-1	1
-2	0,25



→ $f(x) = x$

x	y
0	0
1	1
-1	-1

→ $g(x) = \frac{x}{8}$

x	y
0	0
8	1
1	8

Ebakets puntuak

• $f(x)$ eta $g(x)$. $x = \frac{x}{8}$ $\frac{x}{8} - x = 0$
 $-7/8 x = 0$

• $f(x)$ eta $h(x)$

$$\begin{cases} y = x \\ y = 1/x^2 \end{cases}$$

$$x = \frac{1}{x^2}$$

$$x^3 = 1 \quad \boxed{x=1 \quad y=1}$$

• $g(x)$ eta $h(x)$

$$\begin{cases} y = x/8 \\ y = 1/x^2 \end{cases}$$

$$\frac{x}{8} = \frac{1}{x^2}$$

$$x=2 \quad y=0,25$$

$$A = \int_0^1 (f(x) - g(x)) dx + \int_1^2 (\underline{h(x)} - g(x)) dx =$$

$$= \int_0^1 \left(x - \frac{x}{8}\right) dx + \int_1^2 \left(\frac{1}{x^2} - \frac{x}{8}\right) dx =$$

$$= \int_0^1 \frac{7x}{8} dx + \int_1^2 \left(x^{-2} - \frac{x}{8}\right) dx =$$

$$= \frac{7}{8} \frac{x^2}{2} \Big|_0^1 + \frac{x^{-1}}{-1} - \frac{x^2}{16} \Big|_1^2 = \frac{7}{16} + \left(-\frac{1}{2} - \frac{4}{16}\right) - \left(-1 - \frac{1}{16}\right)$$

$$= \frac{7}{16} + \frac{-8-4}{16} - \frac{-16-1}{16} = \frac{7}{16} + \frac{-12}{16} + \frac{17}{16} = \frac{7-12+17}{16} = \frac{12}{16}$$

$$= \boxed{\frac{3}{4} u^2}$$